**Nicole Zhou**
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## EDUCATION

**Cornell University, College of Engineering**, Ithaca, NY — Expected May 2029
Intended Bachelor of Science in Electrical and Computer Engineering and Minor in Physics — GPA: 3.96
Coursework: Digital Logic, Embedded Systems, Data Science, Nanoscience and Nanoengineering
**Interlake High School**: GPA: 4.0/4.0, IB: 42/45, SAT: 1600, U.S. Presidential Scholar Candidate — *Bellevue, WA*

## EXPERIENCE

**Cornell Autonomous Sailboat Project Team** — October 2025 – Present
*Navigation Electrical Subteam* — *Ithaca, NY*

- Redesigning main controls PCB and implementing non-intrusive power monitoring system to measure per-subsystem consumption and support integration of solar power in experimental research-oriented trans-Atlantic boat
- Designed custom PCB in KiCad and wrote firmware that enables boat status monitoring, remote-control latched emergency shutdown, and selective toggle-based power cycling of onboard subsystems via UART

**Computer Systems Laboratory** — December 2025 – Present
*Napp Lab Researcher* — *Ithaca, NY*

- Developed gamepad teleoperation for a Pi 5 block-stacking robot with analog differential drive, polar-coordinate uArm Swift controller, and servo wrist in Python as groundwork for future autonomous construction
- Created an open-source Python library to control DDSM115 servo motors over RS485 serial bus and assembled robot chassis from aluminum T-slot rails with 3D-printed PLA motor and electronics mounts

**Center for Integrated Electronic and Biological Organisms** — June – August 2026
*Computational Modeling Intern at Morsut Lab* — *Los Angeles, CA*

- Built dynamic binarization, quantitative FFT, and autocorrelation analysis tools to extract pattern wavelength, domain size, and nearest-neighbor spacing from Turing patterns in synthetic JAPI-based genetic stem cell circuits
- Prototyped an ESP32-driven LED matrix display for interactive real-time visualization of simulation states

**Fred Hutchinson Cancer Center** — September 2024 – June 2025
*Machine Learning Intern at Dey Lab* — *Seattle, WA*

- Created and optimized a multilayer perceptron model to predict colorectal cancer risk with over 80% accuracy
- Applied principal component analysis and recursive feature elimination to reduce dimensionality in an 84-sample muti-omic dataset with 1,224 metabolites, 36 gut microbiome bacterial taxa, and DNA k-mer frequencies
- Validated that results were statistically significant with SMOTE, k-fold cross-validation, and permutation testing

## PROJECTS

**Automatic Door Monitoring System** — January – May 2026
*ECE 2300 Digital Logic Course* — *Ithaca, NY*

- Implemented a single-cycle RISC-V processor in Verilog on an Intel Cyclone V FPGA with memory-mapped peripherals including a LiDAR sensor polled over SPI to trigger audible alerts and count entries
- Wrote and verified over 5,000 lines of RTL and gate-level Verilog for 50+ hardware modules including an SPI controller, 32-bit ALU, seven-segment driver, and FSM-based tone player

**Wearable Sleeve Pulse Oximeter** — June – August 2024
*MIT Beaver Works Summer Institute* — *Boston, MA*

- Created a pulse oximeter device that integrated electronic components including a MAX30101/32664 PPG sensor and Arduino microcontroller via conductive thread into a custom three-layer single jersey sleeve
- Presented final prototype enabling real-time Bluetooth health metric monitoring and haptic alert system when oxygen saturation dropped below 90% to over 2,000 people both virtually and in-person

## SKILLS AND INTERESTS

**Technical skills:** KiCad, Altium, FPGA, ROS 2, Linux, Git, Docker, PyTorch, UART, Fusion 360, Onshape
**Languages:** C, C++, Python, Verilog, SystemVerilog, Java, JavaScript, Assembly (RISC-V, ARM)
**Campus Involvement**: Cornell Maker Club, IEEE at Cornell, Cornell Computer Repair Association, Outdoor Odyssey